

Flight to the Moon

Y26 (2026) — English translation

In this problem, we will consider orbits that allow a flight to the Moon and a return back. In the simplest cases, such a flight is typically modeled as an elliptical orbit with a major axis equal to r_m , the distance from the Earth to the Moon. However, such an orbit is not optimal or safe. The main disadvantage is that the spacecraft must be captured by the Moon's gravity. During this capture, engines must be used to enter orbit; a failure at this stage could result in a crash on the lunar surface.

This safety issue is addressed by a **free-return trajectory**. The Moon acts as a gravitational slingshot, creating a "figure-eight" trajectory. This was used for the Soviet *Luna-3* probe and for the emergency return of the *Apollo 13* mission.

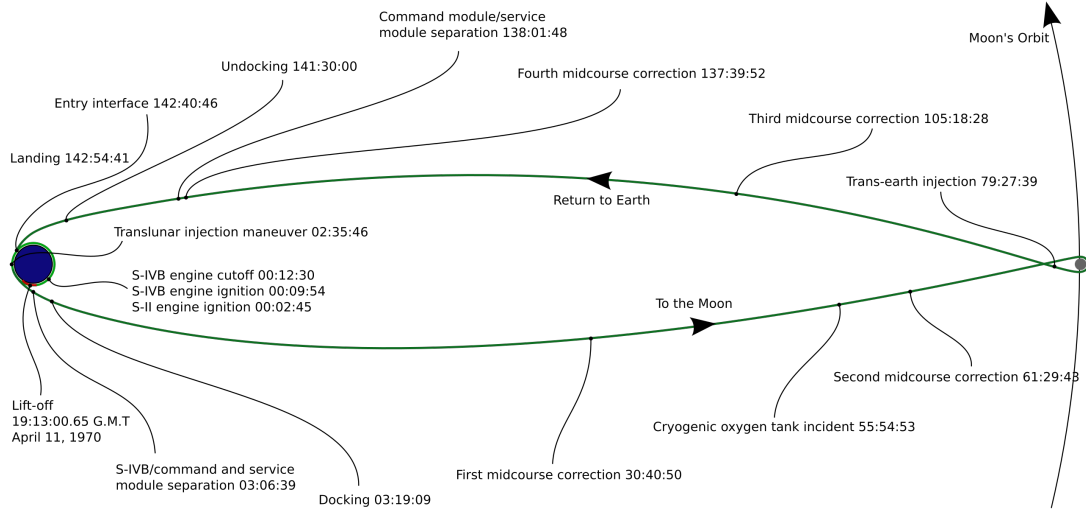


Figure 1: Chronology of the Apollo 13 mission.

The following physical constants may be required:

- Gravitational constant: $G = 6.67 \cdot 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$
- Mass of the Earth: $M_e = 6 \cdot 10^{24} \text{ kg}$
- Mass of the Moon: $M_m = M_e/81$
- Radius of the Earth: $R_e = 6400 \text{ km}$
- Radius of the Moon: $R_m = 1700 \text{ km}$
- Earth-Moon distance: $r_m = 3.85 \cdot 10^8 \text{ m}$
- Orbital period of the Moon: $T_m = 27.4 \text{ days}$
- Orbital speed of the Moon: $v_m = 2\pi r_m/T_m$

Part A. Takeoff from Earth (3.0 points)

Most of a rocket's mass is first-stage fuel required to reach orbital velocity and overcome the atmosphere. We will consider a simplified vertical takeoff along the y -axis. The rocket has initial mass m_0 and is subject to gravity and drag $\vec{F} = -k v \vec{v}$. It expels gas at a constant mass flow rate μ with relative speed u relative to the rocket.

Numerical values: $g = 9.8 \text{ m/s}^2$, $\mu = 2700 \text{ kg/s}$, $u = 4.5 \text{ km/s}$, $k = 0.25 \text{ kg/m}$ (near surface), $a = 6600 \text{ km}$ (orbital radius), $m_0 = 500 \text{ T}$.

A1	Write the expression for $\ddot{y}$ in terms of $\dot{y}$, k , m_0 , μ , u , g , and t .	0.50
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A2	Neglecting drag, find the speed $v(m)$ and coordinate $y(m)$ at the moment the mass is m . Express in terms of m , g , u , μ , and m_0 .	1.00
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A3	Assuming k is constant with height, find numerically the height H_c where the drag force equals the force of gravity.	0.80
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A4	Estimate the ratio β_1 of initial to final mass required to reach circular orbital speed v_a at radius a . Use $\beta_1 = 15$ for later parts.	0.70
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Part B. Free Return Trajectory (6.0 points)

B1	Find the distance r_c from the Moon where the gravitational forces of the Earth and Moon are equal in magnitude.	0.30
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The sphere of radius r_c is the "Lunar zone of attraction". Inside, we only consider Lunar gravity ($\Delta U = \Delta U_m$); outside, only Earth gravity ($\Delta U = \Delta U_e$). The ship starts from a circular orbit of radius a , accelerating to v_0 . Let b be the minimum distance to the Moon's center.

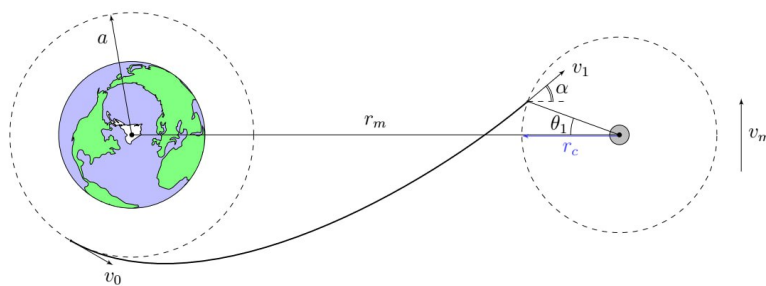


Figure 2: Trajectory geometry.

B2	In the Moon's frame, write the specific angular momentum L_0^m and speed v_2 at distance r_c . Use $s = \sin(\alpha + \theta)$, $s' = \sin \alpha$, and $c = \cos \theta$.	0.50
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B3	Using energy conservation in the Moon's frame, find v_1/v_m in terms of s , s' , c , b/r_c , and v_{II}^m/v_m .	1.20
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B4	Expand v_1 to first order in b/r_c to show $v_1 \approx \frac{v_m c}{s} + B_0 \frac{b}{r_c}$. Express B_0 .	1.00
B5	Using $\alpha \ll 1$, show that $v_1 \approx \frac{v_m}{\tan \theta} + A\alpha + B \frac{b}{r_c}$. Express A and B .	0.60
B6	In the Earth's frame, find α to first order in r_c/r_m and a/r_m in terms of $a, r_c, r_m, v_1, v_0, \theta$.	0.60
B7	Neglecting terms of order a/r_m and r_c/r_m , express v_0 in terms of v_1 and v_{II}^e .	0.40
B8	For a symmetric orbit, use the conic section equation to find an equation for θ using $v_1, \alpha, v_m, G, M_m, b, r_c$.	1.00
B9	Find β_2 , the mass ratio for acceleration from v_a to v_0 . Calculate its value using data from Part A.	0.40

Part C. Lunar Landing (1.0 points)

Assume a close approach with $b = 1800$ km. Since $h = b - R_m \ll R_m$, fuel is only used to brake to zero speed at the pericenter.

C1	With $v_2 = 1.5$ km/s, find the mass reduction factor β_3 during landing.	0.50
C2	Find the initial mass m required to deliver a 40 ton cargo to the Moon.	0.50